

SITKA, AK, USA

WMO: 703710

Lat: 57.048N

Lon: 135.365W

Elev: 14

StdP: 14.69

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 25333

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	17.7	21.3	-1.9	5.0	21.1	3.1	6.5	24.9	28.5	43.2	25.7	42.5	6.5	90	0.654

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.0	66.7	58.4	64.1	57.5	62.2	56.7	60.2	64.5	58.9	62.6	57.9	61.0	7.8	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.2	72.4	61.4	57.1	69.6	60.2	56.1	67.1	59.4	26.5	64.8	25.6	62.5	24.9	61.0	75.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.8	21.0	18.7	DB	14.2	73.9	4.2	4.3	11.2	77.0	8.7	79.6	6.4	82.0	3.3	85.1	(4)
(5)			WB	11.7	62.8	4.4	3.4	8.5	65.2	5.9	67.2	3.4	69.1	0.2	71.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	45.5	36.8	36.6	37.7	42.6	48.0	53.0	56.4	57.4	53.4	46.6	40.1	37.3	(6)
(7)		DBStd	9.03	6.91	5.84	5.55	4.08	4.03	3.36	2.79	2.93	3.36	4.23	5.79	6.18	(7)
(8)		HDD50	2306	409	375	383	226	87	7	0	0	9	119	298	393	(8)
(9)		HDD65	7106	874	795	847	672	528	361	267	237	349	571	747	858	(9)
(10)		CDD50	677	1	0	1	4	25	96	198	229	109	13	1	0	(10)
(11)		CDD65	1	0	0	0	0	0	0	0	1	0	0	0	0	(11)
(12)		CDH74	8	0	0	0	0	0	2	2	4	0	0	0	0	(12)
(13)		CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	7.6	8.8	8.0	8.3	7.6	6.9	6.2	5.8	5.8	6.7	8.4	9.2	9.5	(14)
(15)	Precipitation	PrecAvg	87.3	8.1	6.3	6.0	4.9	4.6	3.7	4.1	6.7	11.3	13.9	9.3	8.4	(15)
(16)		PrecMax	123.0	15.7	15.6	12.9	11.2	10.0	8.1	9.5	19.5	20.7	30.8	16.8	15.1	(16)
(17)		PrecMin	60.1	1.6	0.2	0.9	1.1	1.2	0.9	1.7	1.7	3.9	6.3	1.7	2.2	(17)
(18)		PrecStd	14.7	3.5	3.6	2.6	2.4	2.3	2.0	1.8	3.9	4.6	5.0	3.7	3.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.6	52.3	54.2	61.5	67.9	69.3	68.4	71.2	67.7	58.7	54.5	52.1	(19)
(20)			MCWB	46.5	43.4	45.0	49.2	51.0	58.8	60.5	61.6	57.9	52.8	47.1	45.3	(20)
(21)		2%	DB	48.6	48.1	49.5	54.2	60.0	63.3	64.8	66.6	62.9	55.8	51.7	49.5	(21)
(22)			MCWB	43.7	42.1	42.9	46.1	49.9	56.3	58.6	59.6	55.4	51.5	46.4	44.5	(22)
(23)		5%	DB	46.8	46.1	46.9	50.9	56.5	60.7	62.6	64.3	60.7	54.1	49.8	47.2	(23)
(24)			MCWB	42.5	41.4	41.8	44.1	49.8	54.9	57.5	58.7	55.3	50.0	45.6	43.1	(24)
(25)		10%	DB	45.1	44.5	45.0	48.8	54.1	58.3	60.9	62.4	58.7	52.6	47.8	45.3	(25)
(26)			MCWB	41.5	40.6	40.8	43.2	49.0	53.5	56.7	57.7	54.6	48.8	44.4	42.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.1	45.8	46.5	50.6	54.0	60.0	61.3	62.8	59.8	55.1	50.2	47.8	(27)
(28)			MADB	51.4	48.3	51.2	58.9	61.1	66.3	66.7	67.7	63.1	56.4	51.9	49.8	(28)
(29)		2%	WB	44.9	43.5	44.5	47.1	52.1	57.5	59.4	60.8	58.0	53.0	47.9	45.6	(29)
(30)			MADB	47.1	46.5	47.9	52.2	57.3	61.9	63.7	64.9	60.7	54.6	50.2	48.0	(30)
(31)		5%	WB	43.3	42.2	43.0	45.7	50.9	55.8	58.2	59.4	56.7	51.3	46.2	43.8	(31)
(32)			MADB	45.6	45.2	45.9	49.6	55.3	59.8	61.7	63.1	59.2	53.0	49.0	46.7	(32)
(33)		10%	WB	42.1	41.0	41.4	44.2	49.6	54.2	57.2	58.4	55.4	49.9	44.6	42.2	(33)
(34)			MADB	44.5	44.0	44.5	48.0	53.5	57.3	60.2	61.5	57.9	51.9	47.5	45.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.3	8.2	8.9	9.9	9.2	8.2	7.0	8.0	8.4	8.0	7.6	7.3	(35)
(36)			MADB	7.5	8.7	11.2	13.7	14.5	12.7	10.5	12.3	12.2	9.0	7.9	7.5	(36)
(37)			MCWBR	4.9	5.2	7.0	7.8	8.0	7.1	6.1	7.0	7.4	6.3	5.7	5.3	(37)
(38)		5% WB	MADB	6.9	7.7	9.7	12.4	12.2	11.3	9.2	10.4	9.4	7.0	7.4	7.5	(38)
(39)			MCWBR	5.4	5.5	6.8	7.7	7.3	6.8	5.8	6.4	6.6	5.9	6.3	6.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.301	0.313	0.332	0.396	0.374	0.371	0.382	0.362	0.356	0.340	0.305	0.282	(40)	
(41)		taud	2.249	2.386	2.374	2.186	2.380	2.438	2.445	2.524	2.496	2.394	2.271	2.190	(41)	
(42)		Ebn at Noon	180	237	262	257	274	276	270	269	253	221	176	151	(42)	
(43)		Edh at Noon	16	22	28	39	35	33	32	28	25	21	16	13	(43)	
(44)	All-Sky Solar Radiation	RadAvg	120	337	688	1155	1465	1505	1272	1098	706	374	162	86	(44)	
(45)		RadStd	18	65	99	113	157	135	150	139	118	50	24	14	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	+2.50	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)		N/A	+2.00	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page